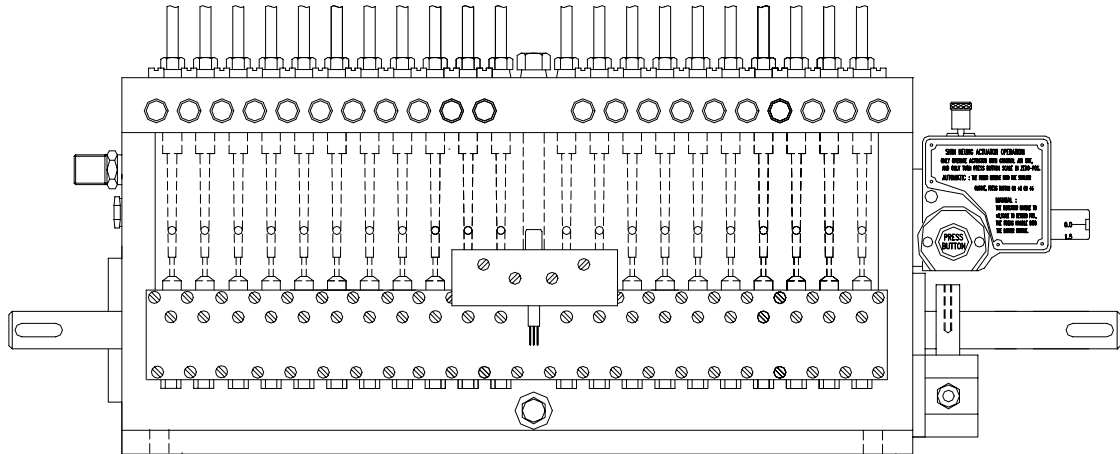


ATLAS LUBRICATOR



MANUAL

- Engine no. : _____
- Engine type : 7S50MC - C
- Engine builder : _____
- Shipyard : _____
- Hull no. : _____
- Pump plunger diameter(mm) : 4.5mm
- No. of Lub. points per lubricator : 21
- Alarm unit : 15F001/010
- Stroke change per turn of adjusting bolt(mm) : _____
- Special features : _____

SH SHIN HEUNG PRECISION CO., LTD.

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ATLAS CYLINDER LUBRICATOR

Before use and/or interference in the lubricator and/or connected functions please read the entire manual.

CONTENTS

Lubricating Unit

1. Function of cylinder lubricator	A1
2. Structure of cylinder lubricator	A1
3. Adjustment of lubrication timing	A2
4. Adjustment of lubricating oil quantity	A2
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Load Change Dependent(LCD) Lubrication

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2. Function	L1
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Alarm sensor

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El-Heater(Optional)

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Lubricating unit

1. Function of cylinder lubricator

The cylinder lubricator is a specific volumetric pump with a plunger and barrel. The cylinder lubricator is designed to deliver a specified quantity of lube-oil to the cylinder liner.

The main purpose of a cylinder lubricator is to reduce corrosion and friction and also to increase the sealing effect between the piston ring and the cylinder liner.

The function of the lubricator can be monitored by non-flow and/or low level alarm devices mounted on the lubricator.

2. Structure of cylinder lubricator

The lubricator is connected to the cylinder oil supply tank.

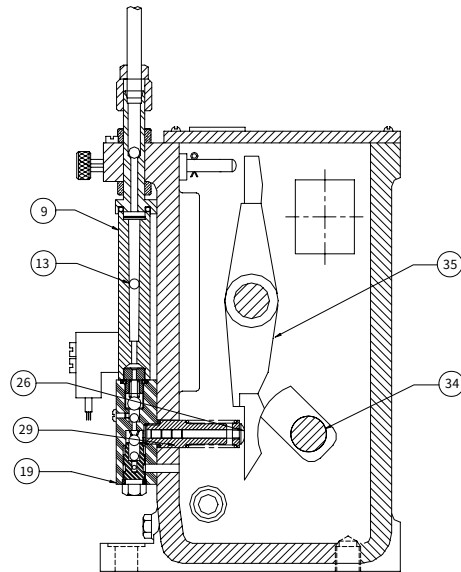
The cylinder lubricator is a form of reciprocating plunger pump, the components of which are shown in the enclosed drawings 1-100-A, 1-101-A, 1-102-A, 1-103-A, 1-104-A, 1-105-A.

The drive mechanism, which is connected to the engine chain drive shaft via coupling, rotates a cam shaft(34) and thereby the levers(35) push the plungers (26).

Plunger bushes(29) are filled with a certain amount of cylinder lube-oil through holes in the pump body(19) when the plungers(26) are pulled.

Then cylinder lube-oil is fed to the cylinder liner via the pump body(19). The balls(13) come upwards in indicators(9) while the levers(35) push the plungers(26).

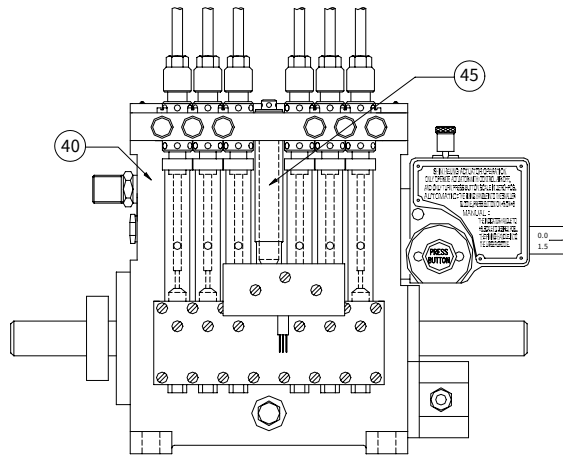
The steel balls(13) are installed inside of tapered indicators(9). These balls are set to come up or down to indicate a certain height. This height is dependent on the rates of the pumped lube-oil which is proportional to the plunger stroke.



The oil level in the oil case(40) may be estimated by looking at the level glass(45).

Theoretically, the oil feed rates should be nearly equal to all the points if the measured height of the steel balls are equal to each lubricating point.

But, in practice, it is not suitable for measuring the lubricating oil consumption directly because a viscosity and a specific gravity of the oil are dependent on the temperature.



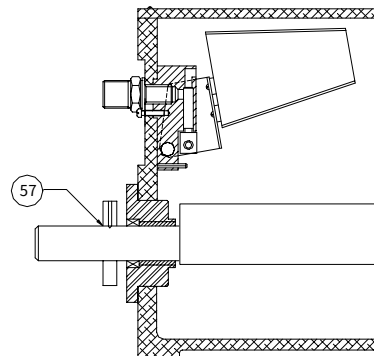
The cylinder lubricator oil case(40) is automatically filled with lube-oil to a certain height by a float valve complete.

3. Adjustment of lubrication timing

Lubrication timing is to be adjusted by means of timing gauge which is supplied by the engine maker.

A special mark("A") is stamped on the timing plate(57) for the adjustment of the lubrication timing.

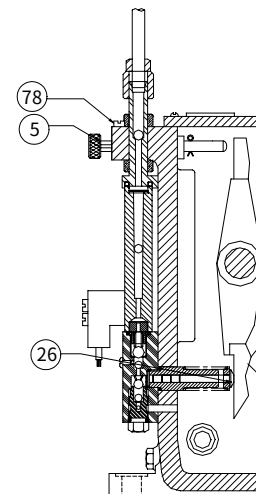
While the piston moves, lubrication is completed normally before the piston ring passes the lubricating point in the cylinder liner.



4. Adjustment of lubricating oil quantity

Lube-oil quantity can be adjusted by increasing or decreasing the stroke of the plungers(26) which can be changed by the adjusting bolts(5).

The set screws(78) above the adjusting bolts(5) should be loosen before turning the adjusting bolts. The required stroke can be gained by turning the adjusting bolts(5) clockwise to decrease the stroke or counter-clockwise to increase the stroke.



Two revolutions of the adjusting bolt gives a stroke change of 1mm. The set screws(⑧) should be tighten into the key way of the adjusting bolts(⑤) after the adjustment of the plunger stroke. The action of the adjusting bolt is proportional to changes in the plunger stroke.

The optional injection at each of the lubricating points is carried out through the adjustment of the plunger stroke using the above mentioned adjusting bolts(⑤).

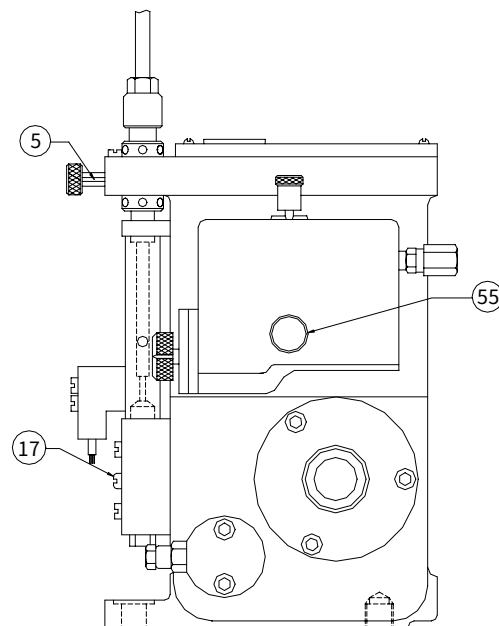
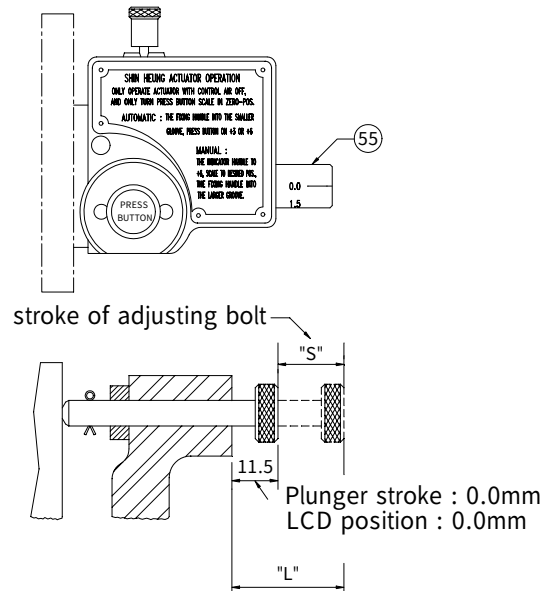
It should be noted that the indicator handle(⑤) which is fitted right on the cylinder lubricator must be set at the 0.0 position during the adjustment of the plunger stroke.

An increase or decrease by 0.5mm per one revolution at length 'L' of the adjusting bolts corresponds to an increase or decrease of the plunger stroke by approximately 0.5mm.

When the engine has been idle for a long period of time or if any part of the pump has been replaced with a new part, it may happen that the remaining oil may not be pumped at all or only small amount of oil may be discharged due to accumulated air in the pump. In such cases, it is necessary to bleed the pump manually.

Unscrew the air vent screws(⑪) and pump the adjusting bolts(⑤) manually until the air has been removed from the pump completely. Then keep the adjusting bolts in that position while reinserting and fastening the air vent screws(⑪).

If the lubricator is in a position where pump action is difficult to perform manually because the cam shaft presses on the plungers, you can turn the engine with the engine turning gear in order to release the plungers.



The indicator handle(55) regulates the stroke of all pumps in the lubricator at the same time. If it is necessary to increase the oil feed rate temporarily during maneuvering or running-in of the piston ring and cylinder liner, the indicator handle(55) is used. The indicator handle is set to the position 0.0 in the normal operation.

The handle in turn is set in steps from the maximum feed rate at the position of 6.0. If it is necessary to decrease the feed rate, use the handle to decrease the feed rate by moving it to the positions of 4.5, 3.0, 1.5, 0.0.

If the handle is moved to the position 6.0, the stroke will be increased by 6mm or 4 steps as each step makes 1.5mm increase. If the handle is moved from 0.0 to 3.0, the increase of each plunger stroke is 3mm or 2 steps. (See page A-15, Dwg. No. BL006)

Warning :

- **Do not open the lubricator during operation.**
- **Avoid touching any moving parts of the lubricators during operation, as this may result personal injury or damage to the parts.**
- **Do not disassemble the float valve during operation, as a loose float valve in an operating lubricator will result in severe damage.**

5. Theoretical cylinder oil pump feed rate.

We suggest the following formula for calculating the pump feed rate per cylinder per day. The formula may also be used for adjusting the stroke or calculating the required stroke setting based on the engine maker/designer recommendation about the guiding feed rate.

$$Q = \frac{3.14 \times d^2}{4} \times S \times E \times N \times n \times 60 [\text{min} / h] \times 24 [h / \text{day}] \times \left(\frac{G}{1000000} \right)$$

Q : Theoretical lube-oil consumption per cylinder(Kg/cyl/day)

d : Plunger diameter (mm)

S : Plunger stroke (mm)

E : Volumetric efficiency of lubricator (=0.98)

N : RPM of engine

n : Number of lub. points

G : Specific gravity of lubricating oil

(Average for SAE 50 cylinder oil : 0.94Kg/liter)

6. Theoretical specific cylinder oil consumption

Theoretical specific cylinder oil consumption is as follows;

$$F = \frac{Q}{P}$$

F : Theoretical specific cylinder oil consumption(g/BHP. h)

Q : Theoretical lubricating oil consumption per cylinder(g/cyl. h)

P : Max. power per cylinder (BHP/cyl)

7. Maintenance

Every time the engine cylinders are overhauled, disassemble each lubricator as well to remove any oil sludge that is deposited in it and to overhaul the valve sections in the oil supply lines.

Also, disconnect the cock connected plug for the floating valve and clean indicators. When reassembling the indicators, be sure to locate it at the upside of the larger tapered correctly.

After reassembling, supply new oil and continue air venting and priming by the adjusting bolts to lubricate the sliding surface of each of the cylinders.

During reassembly, replace all gaskets and seals, and be careful not to get dirt into the systems.

Visually check that oil is supplied through each of the oil feed ports. Check it carefully for any oil leaks through the packings or the joints. If leakage occurs, it may cause a lack of the feed quantity to the cylinder sliding surface and result in the piston seizing up.

Re-tighten the screws in the lubricator within the first three months or after 2,000 hours of normal operation.

Disassemble the lubricators once a year or at suitable intervals.

- ① Clean sludge from the oil case.
- ② Remove and clean the valves in the pump body.

※ When you disassemble the pump body, do so slowly as the plunger might be separated from the spring.

- ③ Clean the indicators.
- ④ Clean the inside of the air vent screws with clean washing oil.

- Do not use cleaning fluids containing alcohol.

After reassembling, refill with oil.

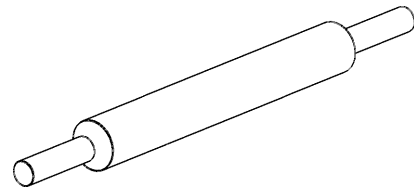
※ For ordering spare parts, refer to the relevant part's drawing and it's associated position number.

In case of certain changes and/or modifications to our products without consultation with us, or the use of components not produced by us, we cannot be held responsible for the operation and function of the lubricators and any resulting damage. Further instructions are available on request.

Continuous development brings changes to our products, so this manual may not be up to date.

8. Special tool

For ATLAS 1, ATLAS 11, we provide the special tool that disassemble or reassemble the lock nut(③) and the gauge support(④).
(See Dwg. No. 1-100-A, Dwg. No. 1-104-A)



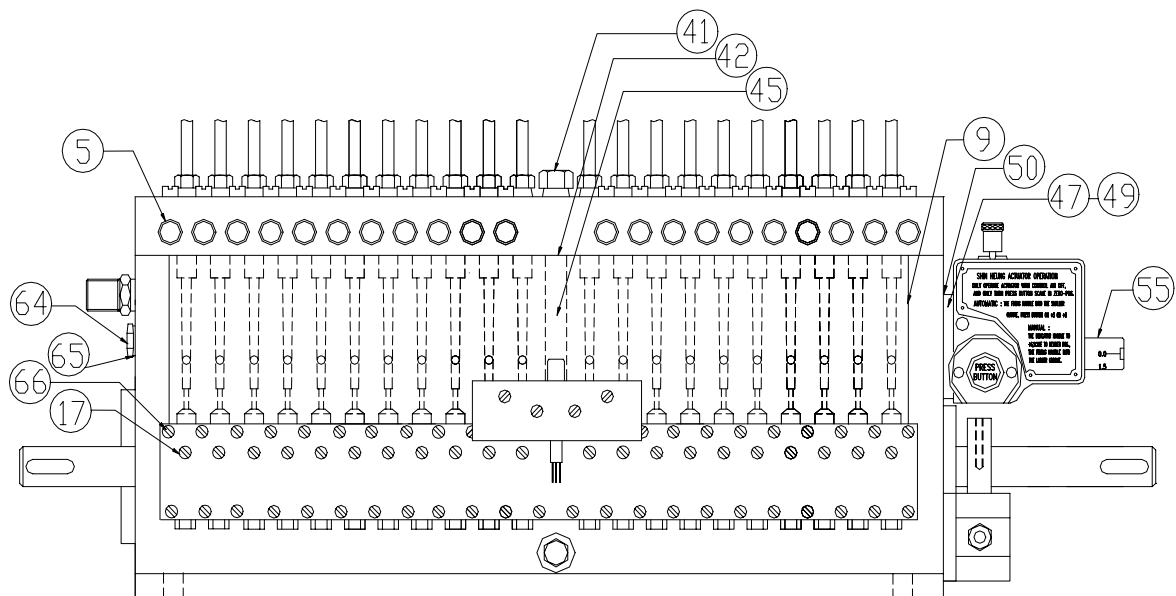
(2 Pcs per Engine)

※ *All of the spare parts are available.*

Spare parts for ATLAS lubricators - Lubricating unit

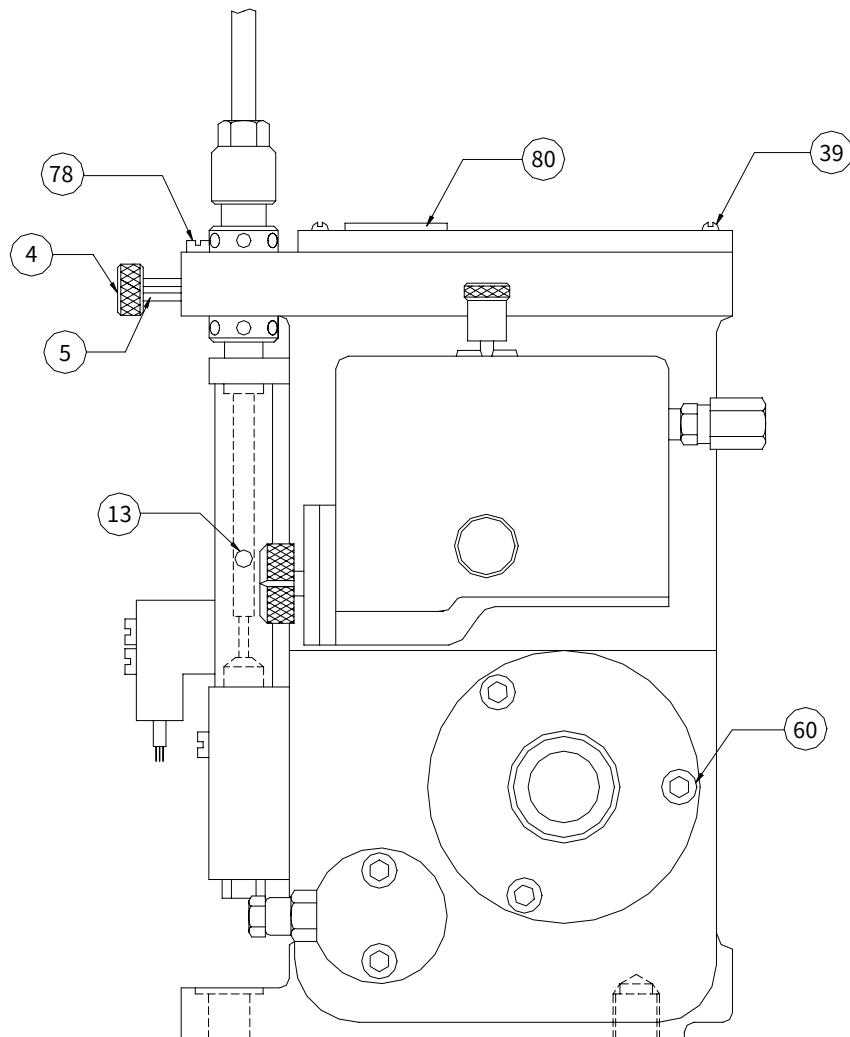
Dwg. No. 1-100-A

Pos. No.	Description
5	Adjusting bolt
9	Indicator
17	Air vent screw
41	Gauge support
42	Packing for gauge support
45	Level glass
47	Flange
49	Oil seal
50	Flange packing
55	Indicator handle
64	Plug
65	Packing for plug
66	Cap screw



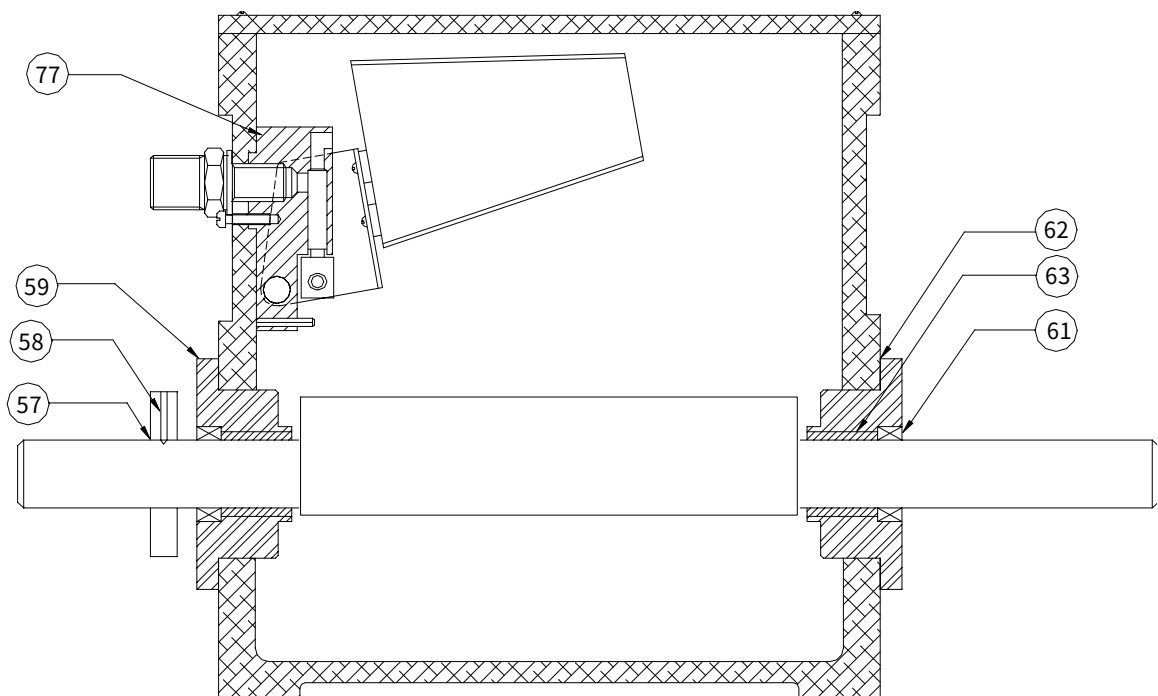
Dwg. No. 1-101-A

Pos. No.	Description
4	Adjusting knob
5	Adjusting bolt
13	Ball for indicator
39	Screw for cover
60	Cap screw
78	Set screw
80	Plug for cover



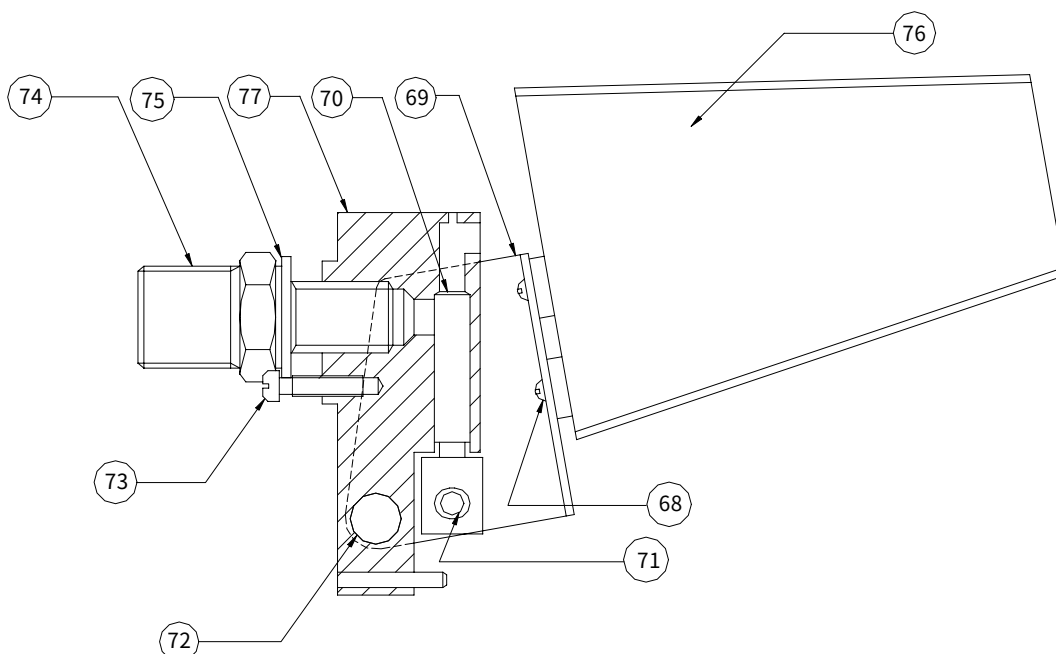
Dwg. No. 1-102-A

Pos. No.	Description
57	Timing plate
58	Set screw
59	End bearing
61	Oil seal
62	Packing for bearing
63	Bush for bearing
77	Float body



Dwg. No. 1-103-A

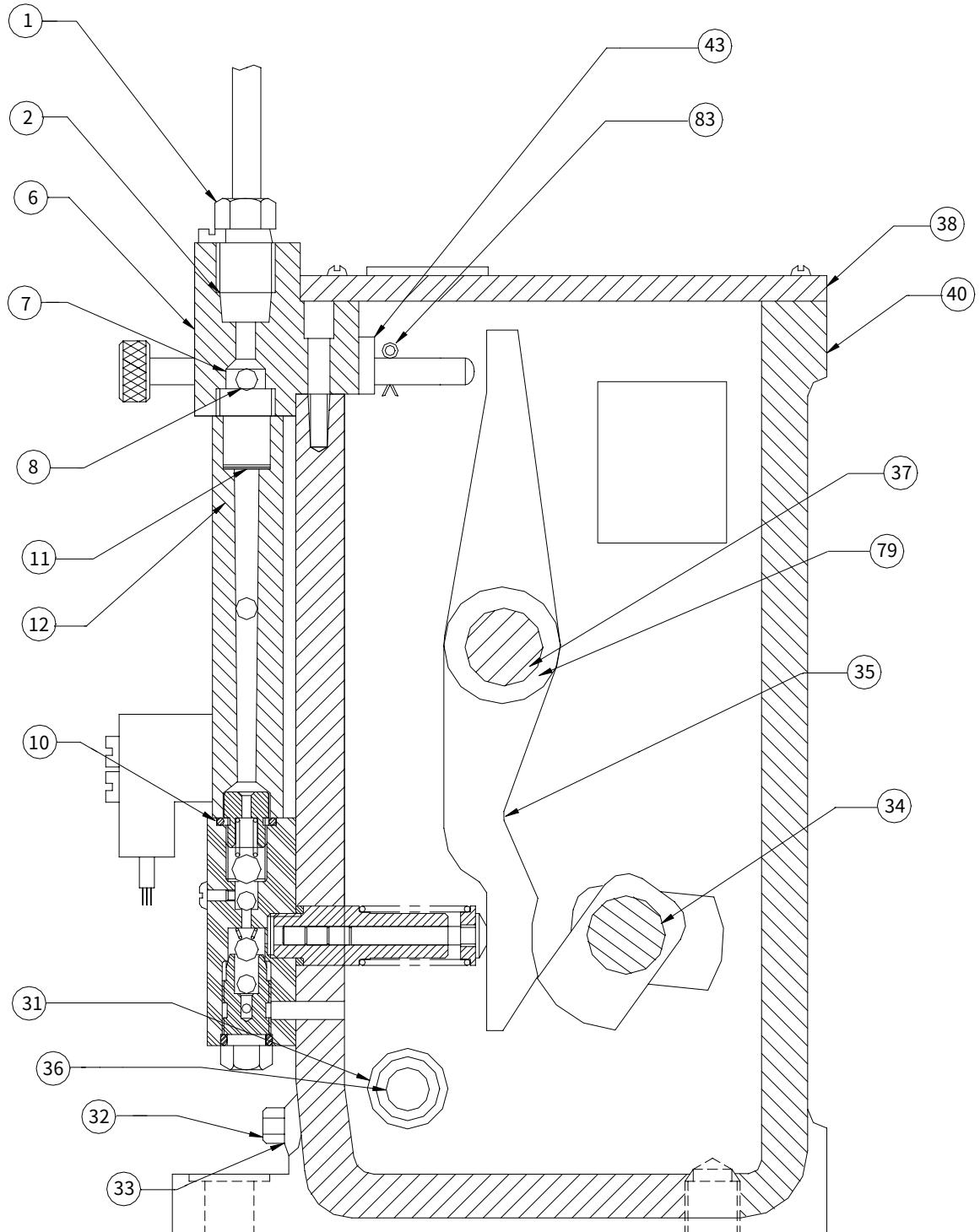
Pos. No.	Description
68	Screw
69	Arm for float valve
70	Valve
71	Pin
72	Pin
73	Float screw
74	Plug for float
75	Packing for float
76	Float
77	Float body



BLANK

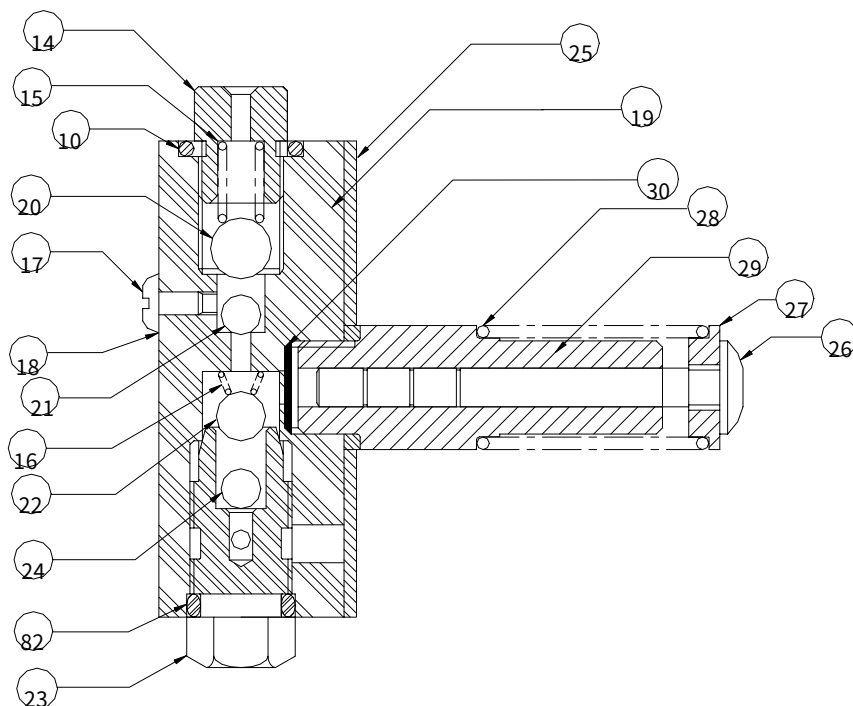
Dwg. No. 1- 104-A

Pos. No.	Description
1	Union nut
2	Sleeve
6	Indicator support body
7	Stopper for ball
8	Ball for valve
9	Indicator
10	O-ring for indicator
11	Stop ring
12	Gasket
19	Pump body
31	El-heater (Option)
32	Drain plug
33	Packing for drain plug
34	Cam shaft
35	Lever
36	Pocket for El-heater
37	Eccentric shaft
38	Cover
40	Oil case
43	Nut for adjusting bolt
79	Distance
83	Split pin



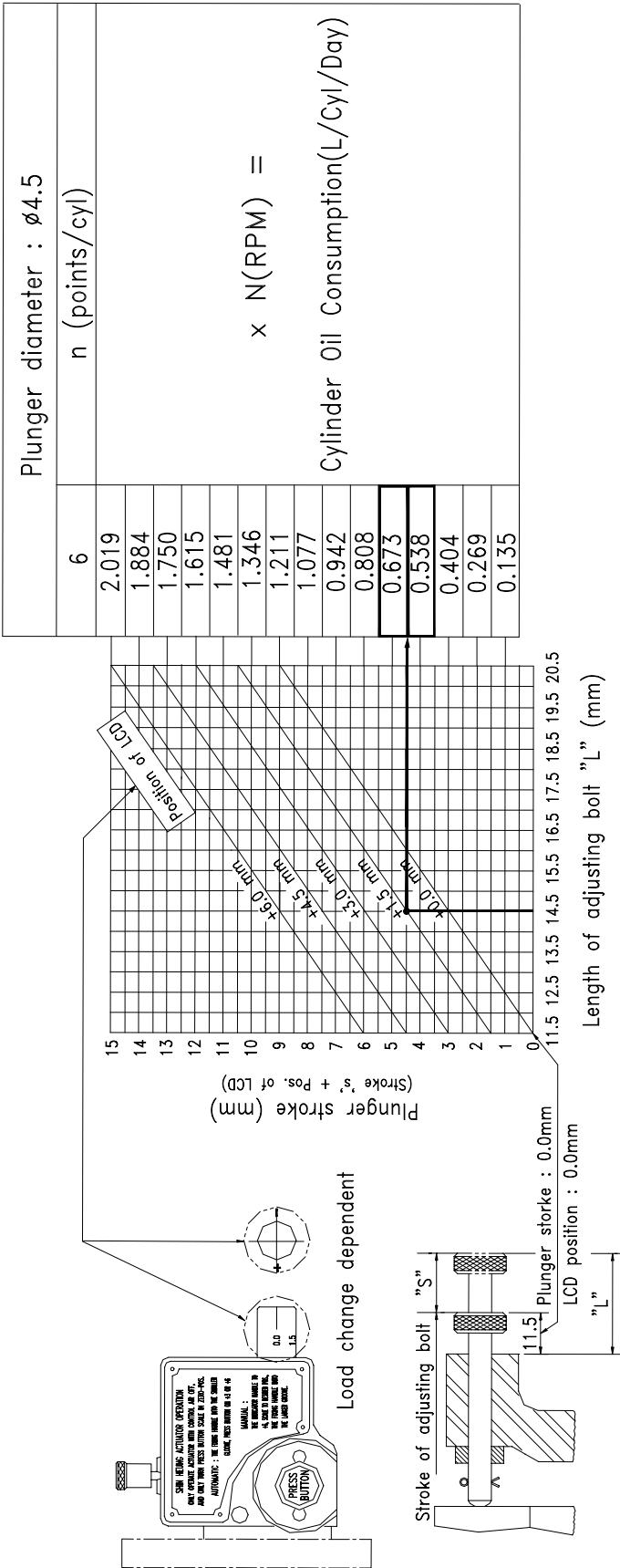
Dwg. No. 1- 105-A

Pos. No.	Description
10	O-ring for indicator
14	Nozzle
15	Discharge spring
16	Discharge spring
17	Air vent screw
18	Packing for air vent screw
19	Pump body
20	Ball for valve
21	Ball for valve
22	Ball for valve
23	Suction valve
24	Ball for valve
25	Gasket
26	Plunger
27	Spring stopper
28	Plunger spring
29	Plunger bush
30	Packing
82	O-ring



”The graph of adjustment for cylinder oil consumption”

※ These data should be taken as a reference due to difference from actual status, such as viscosity of cyl. oil, pump condition etc.



<< Example >>

When, RPM N = 90
 Pos. of LCD: +1.5mm
 Stroke "S" = 3.0mm
 Plunger stroke : 1.5+3.0=4.5mm

N : RPM of engine
 Efficiency of pump : 0.98
 Plunger diameter : $\phi 4.5$
 n (No. of lub. points) : 6

$$Q = \frac{(0.673+0.538)}{2} \times 90 = 54.50 \text{ L/CYL./DAY}$$

SH DWG. NO. BL006

Load Change Dependent(LCD) Lubrication

1. Purpose

The LCD adjustment is recommended to increase the cylinder lube-oil quantity at the start or during the loading operation because the cylinder liner and piston ring are exposed to severe wear.

The lubricator is equipped with LCD which is fitted with a pneumatic actuator for automatic plunger stroke adjustment of the cylinder lube-oil quantity.

This automatically assures an increase in the lubrication during LCD operation.

This improves the engine operation and reduces wear on the cylinder liner and piston ring.

'Load Change Dependent Cylinder Lubrication' automatically provides an extra amount of oil both at engine startup and during changes to the engine load.

2. Function

The cylinder lubricator with LCD is fitted with an actuator which changes not the 'partial lubrication quantity' but the 'overall lubrication quantity'.

This actuator is controlled by a dedicated signal which is generated in the electronic governor.

The control signal is transmitted to the solenoid valve which is actuated by actual load change values bigger than preset values in the engine load.

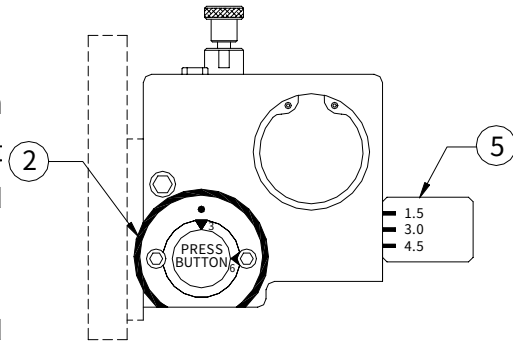
The pneumatic actuator is operated by the solenoid valve which introduces compressed air through the selection of A or B.

The selection of A increases the lubrication quantity. After lubricating for a set time, the conversion to B will result in the adjustment of the stroke by the basic setting.

The adjustment of LCD to the required position is by means of the press button(②) and the indicator handle(⑤).

This mechanism releases an extra lubrication for the set time of 30 minutes both at the start and when change values of the engine load bigger than preset value in the control box.

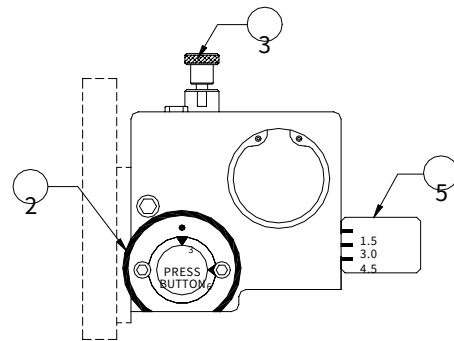
The overall scheme is shown on the enclosed Dwg. No. 2-100-L and on the detail drawings attached to Dwg. No. 2-101-L.



3. Setting

3.1. Adjustment in the Automatic mode by LCD principle

- 1) Disconnect the air supply from the system.
- 2) Lift and turn the fixing handle(③) until the high position.
- 3) Turn and keep the indicator handle (⑤) to the position 0.0.



※ You can only turn the press button, when the indicator handle is at the position 0.0.

- 4) Turn the press button(②) to the '+3' or '+6' position.

**※ press button position "+3" = LCD stroke +3mm
 "+6" = LCD stroke +6mm**

- 5) Reconnect the air supply to the system.

The actuator(s) now operates according to the Load Change principle.

The operation of the lubricator may be checked by observing the indicator handle.

3.2. Adjustment in the specific cylinder

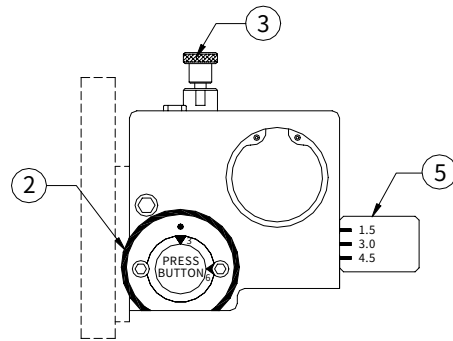
After renewal or repair of a specific cylinder liner and/or specific piston ring, the cylinder lube-oil dosage should be increased. Therefore it is necessary to increase the pump stroke for the specific individual cylinder in the LCD operation.

In order to increase the stroke from 0mm to 6mm, the indicator handle of actuator can be set manually.

1) Disconnect air supply from the system.

2) Turn the press button(②) to the '+6' position

※ Note : The indicator handle(⑤) must be set to the position 0.0.



3) Lift and turn the fixing handle(③), fix to the **high position**.

4) Turn the indicator handle(⑤) to the desired position.

5) Lift and turn the fixing handle(③), fix to the **low position**.

This actuator is now fixed to the desired position.

6) Reconnect the air supply to the system.

Other cylinders except the lubricator of a specific cylinder may still work according to 'Load Change Dependent Lubrication'.

The operation of the lubricator may be checked by observing the indicator handle.

For safety reasons, do not open the actuator during LCD operation. Avoid touching any moving parts of the system as this may result in personal injury or damage to the parts.

4. Maintenance

- Refer to the enclosed drawings 2-101-L. (See Page L-6)

- The actuator is greased through the pinion shaft(21) every fifth year with Aeroshell Grease 7 Mil-G 23827 B or a similar product code 70149.

1) Disconnect the air supply from the system.

2) Disconnect the packing(8) and the nipple(9).

3) Fix the fixing handle(3) to the high position.

Turn and keep 'the press button' to the position "+6".

4) Loosen the adjusting bolt(12) and the adjusting nut(13) completely.

After loosening the screws of the hex bolts(10, 14, 16), separate LCD from the lubricator.

5) Loosen the indicator handle(5) and take out the pinion shaft(21).

Warning :

- During reassembly, the pinion shaft(21) must be set in the original position. If you don't set it to the original position, the indicator handle don't operate correctly. For correct operation, you must use a plier for retaining ring for the stop rings(25).

※ For ordering spare parts, refer to the relevant part's drawing and it's associated position number.

In case of certain changes and/or modifications to our products without consultation with us, or the use of components not produced by us, we cannot be held responsible for the operation and function of the lubricators and any resulting damage. Further instructions are available on request.

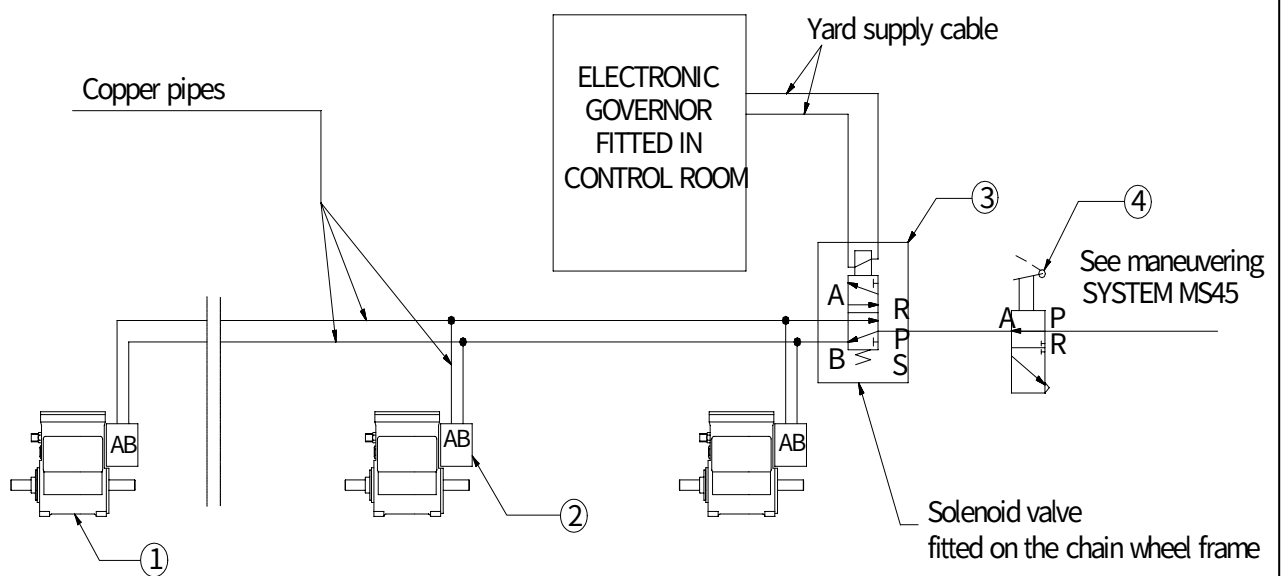
Continuous development brings changes to our products, so this manual may not be up to date.

※ *All of the spare parts are available.*

Spare parts for ATLAS lubricators - LCD

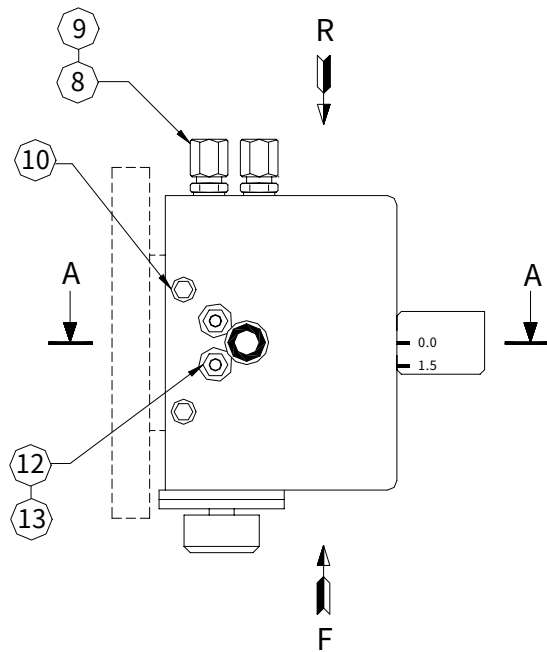
Dwg. No. 2-100-L

Pos. No.	Description
1	Cylinder lubricator
2	Load change actuator
3	Solenoid valve
4	Ball valve

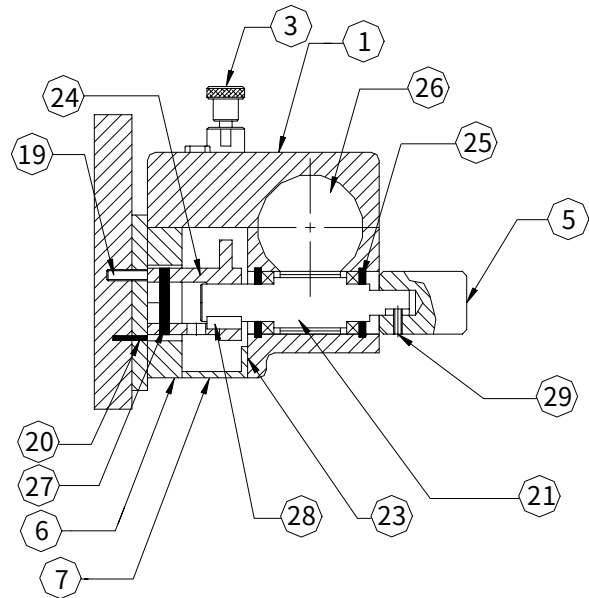


Dwg. No. 2-101-L(Load change dependent assembly)

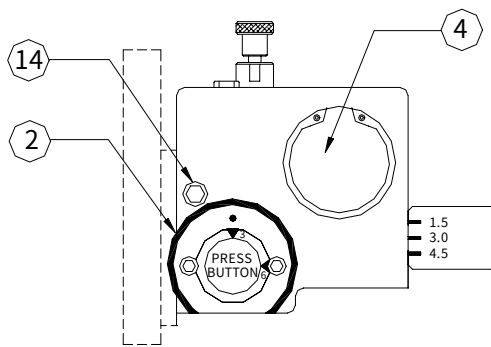
Pos. No.	Description
1	Load change dependent
2	Press button stopper assembly(Ref. Dwg. No. 2-102-L, page L-8)
3	Fixing handle assembly(Ref. Dwg. No. 2-103-L, page L-9)
4	Cover for cylinder
5	Indicator handle
6	Left plate
7	Lower plate
8	Packing
9	Nipple
10	Hex. bolt
12	Adjusting bolt
13	Adjusting nut
14	Hex. bolt
16	Hex. bolt
17	Hex. bolt
18	Stop ring
19	Hex. bolt
20	Spring pin
21	Pinion shaft
23	Spring pin
24	Wing coupling
25	Stop ring
26	Rack piston
27	Spring pin
28	Key
29	Set screw



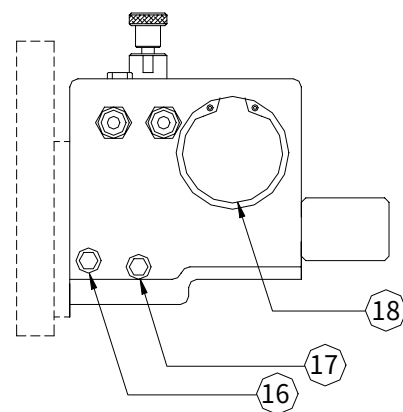
Top view



Section A-A



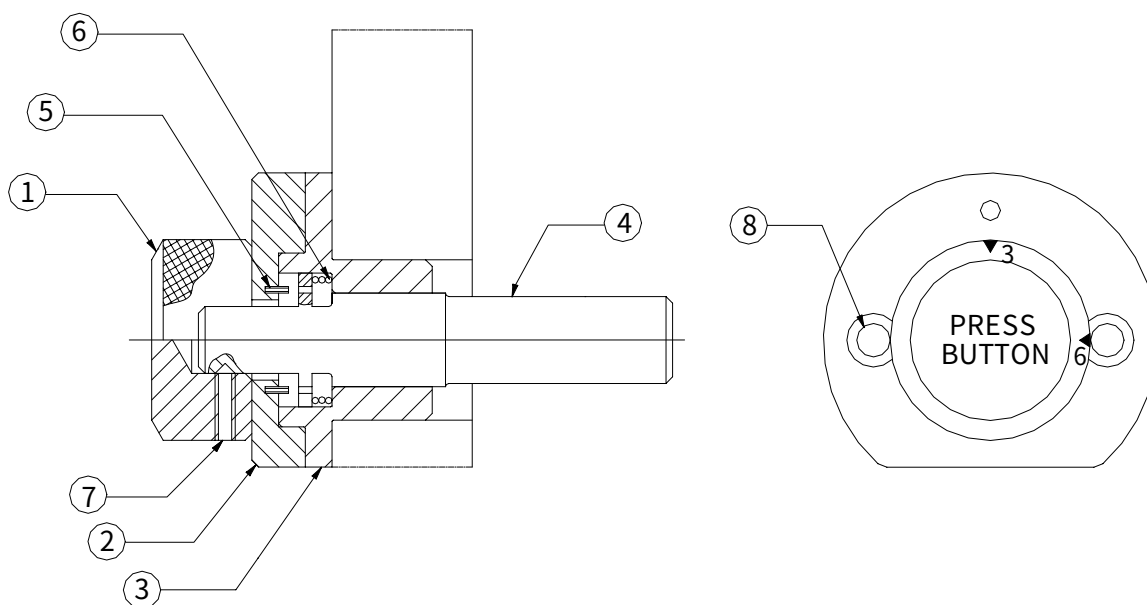
Viewed from F (front side)



Viewed from R (rear side)

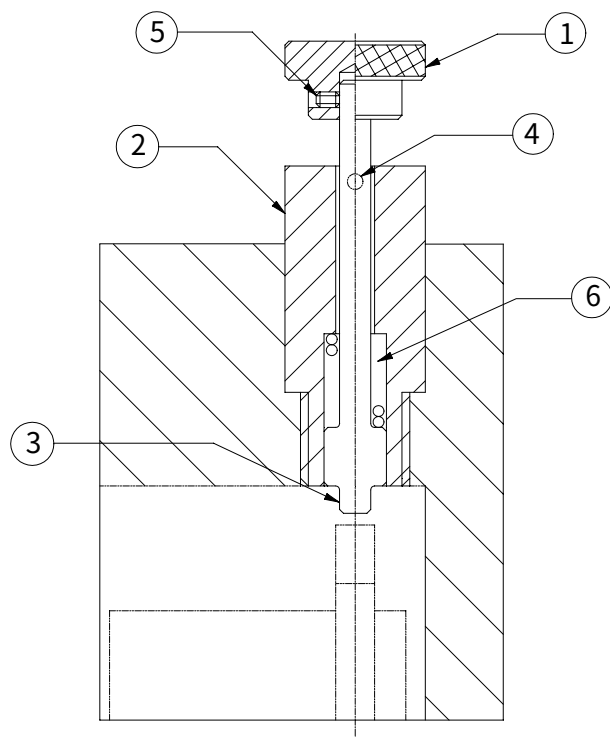
Dwg. No. 2-102-L.(Press button assembly)

Pos. No.	Description
1	Press button
2	Cover
3	Shaft housing
4	Shaft for press button
5	Spring pin
6	Spring
7	Set screw
8	Hex. bolt



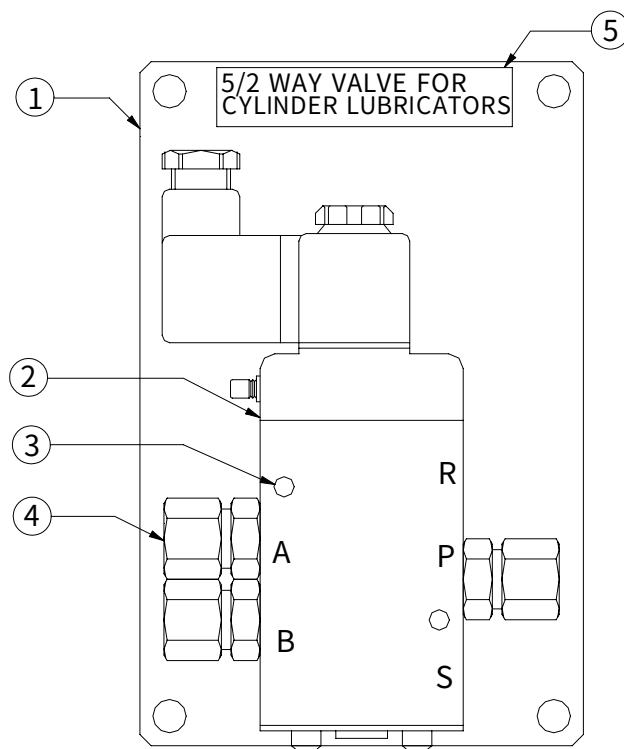
Dwg. No. 2-103-L. (Fixing handle assembly)

Pos. No.	Description
1	Knob
2	Handle
3	Lock pin
4	Spring pin
5	Lock pin
6	Spring



Dwg. No. 2-104-L. (Solenoid valve complete)

Pos. No.	Description
1	Bracket
2	5/2 way v/v complete
3	Hex. bolt
4	Mild stud coupling
5	Name plate



Alarm sensor

1. Function

The sensing unit consists of sensors in front of the indicator and the level glass. The purpose is to make the signal of low level and/or zero flow in the lubricator.

1.1 SU003

Each unit is fitted with 3 sensing units(**1 level : S1 & 2 flows : S2, S2**) and 2 lamps(**1 level and 1 flow**) corresponding to the holes at the bottom of the level glass and the indicator respectively.

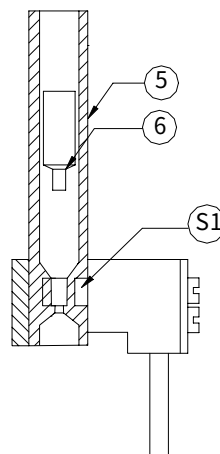
2. Operation

In the indicator with the flow alarm sensor, if the oil flow stops then the ball(④) will fall down and rest at the bottom of the indicator glass(③). Then the alarm sensor will go off.

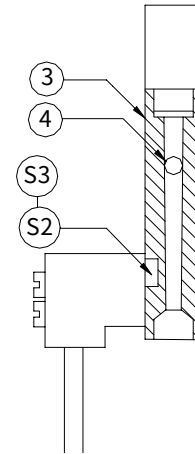
The level float(⑥) of the level glass (⑤) follows the oil level of the oil case.

As Dwg. No. 3-101-S, the system requires a supply voltage of 24V DC $\pm 20\%$.

Low Level Alarm



Non Flow Alarm



When the alarm sounds, the cause must be found and corrected immediately. If the alarm sounds due to a failing oil supply to the lubricator(low level signal), there will usually be time to renew the oil supply before the level at which the pumps begin to suck air and then all operation has to be stopped.

When you inspect the pump unit, check the operation of the alarm before filling the oil or running the pump.

Warning : Do not exceed the max. limits 10W/100V/300mA with ohmic load.

※ If the actual cable color of the alarm sensor is not in accordance with the drawing parts, you must use the corresponding number on the cable to connect it.

※ All of the spare parts are available.

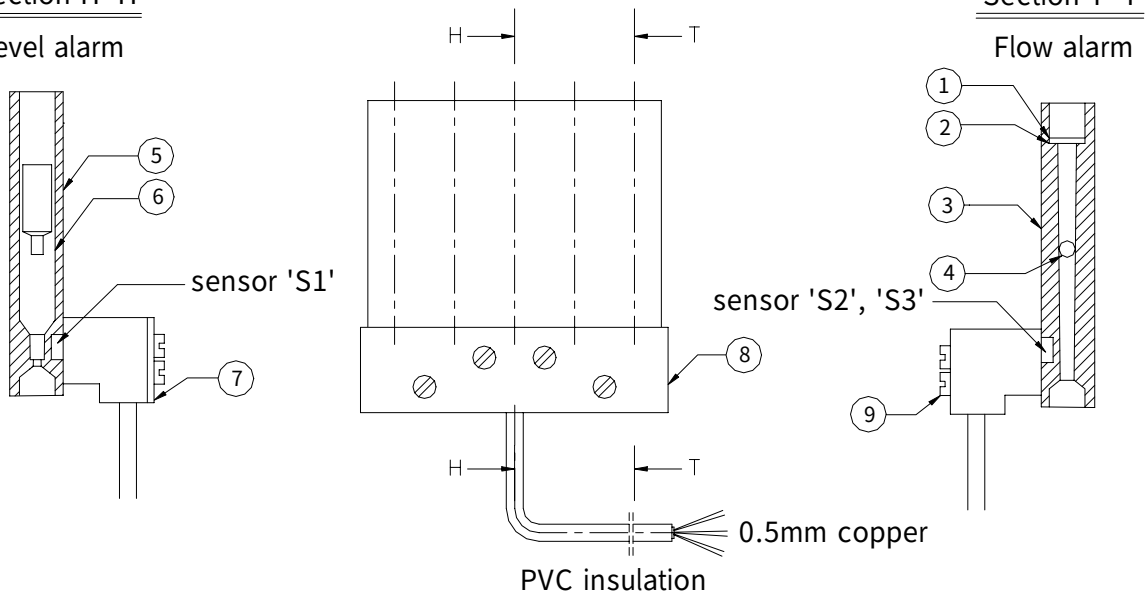
Spare parts for ATLAS lubricators - Sensor SU003

Dwg. No. 3-100-S

Pos. No.	Description
1	Locking ring
2	Fiber ring
3	Indicator
4	Ball
5	Level glass
6	Level float
7	Fitting flange
8	Sensor unit
9	Screw

Section H-H

Level alarm

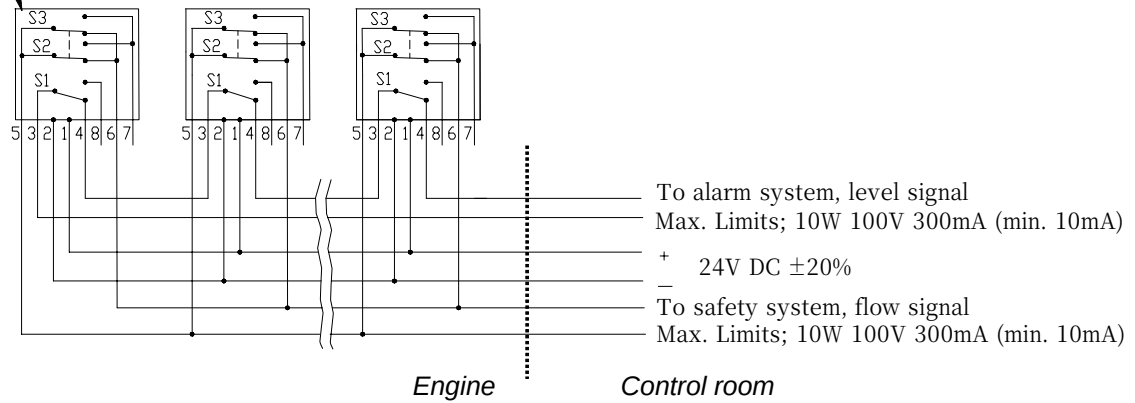


No.	Type	Cable color		Remarks
		15F001	15F010	
1		red	red	×F001 : Open Alarm at Low Level Close Alarm at Zero Flow
2		green	green	
3		blue	blue	
4		yellow	yellow	
5		black	black	×F010 : Open Alarm at Low Level Open Alarm at Zero Flow
6		brown	brown	
7		orange	orange	
8		white	white	

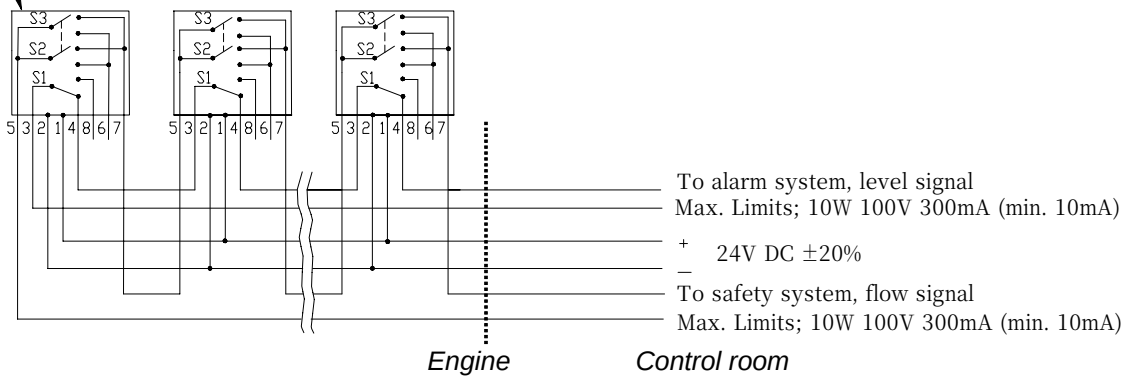
Drawing for ATLAS lubricators - Sensor SU003**Dwg. No. 3-101-S**

※ S1 : Switch for Low Level, S2, S3 : Switch for Non Flow

One cylinder lubricator unit

ATLAS lubricator : **15F001** type

One cylinder lubricator unit

ATLAS lubricator : **15F010** type

Terminal designation corresponds to the color coding or No. on the wire insulation.

The diagram shows the system in the following condition ;
Engine stop position, EI-power on, Zero flow, High level

※ Cable Connection

Function	Normal	Alarm
Low level	3(Blue) —  — 4(Yellow)	3(Blue) —  — 4(Yellow)
	a (option) 3(Blue) —  — 8(White)	a(option) 3(Blue) —  — 8(White)
Non-flow	a 5(Black) —  — 6(Brown)	a 5(Black) —  — 6(Brown)
	b 5(Black) —  — 7(Orange)	b 5(Black) —  — 7(Orange)
Source	1(Red) —  — 2(Green)	

El-Heater (Option)

The lubricator is designed to be mounted with the El-heater. This heater keeps the oil temperature at approx. 40-50°C, or approx. 20°C above surrounding temperature.

The heating facilitates, among other things, the discharge of air from the oil. This is very important for the correct functioning of the lubricator.

Therefore we recommend to mount the El-heater which is connected to one-phase voltage source.

The capacity of voltage(V), power(W) are stamped on the cover of the heater terminal box.

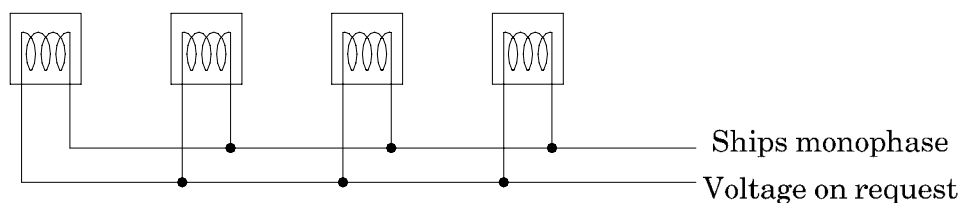
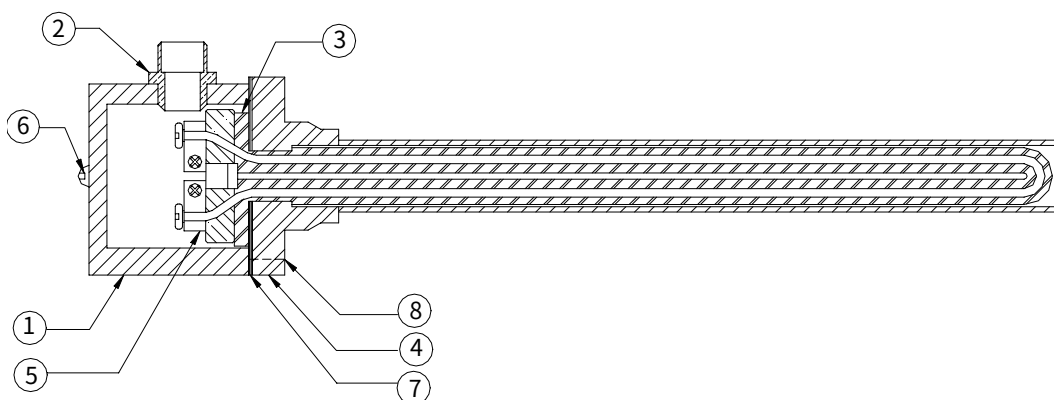
Warning : Before dismounting the El-heater, turn off the power to prevent an electric shock or the ignition of any flammable material.

※ *All of the spare parts are available.*

Spare parts for ATLAS lubricators - EI-heater.

Dwg. No. 4-100-H

Pos. No.	Description
1	Cap for heater
2	Cable gland
3	Flange for heater
4	Case for heater
5	Cartridge heater
6	Bolt for cap
7	Packing for cap
8	Bolt for case



Wire diagram - All types

No. of Lub. points	21
HEATER POWER	100W

The heater ensures oil temperature of 40 ~ 50°C.

All cables and cable connection are to be supplied by shipyard.